Steel decarbonisation timing — handoff for the deck

Investment-timing model, European integrated works · generated 2026-09-23 from a live model run · every figure and number below is produced by the code, not typed in

> **How to use this file.** It is self-contained: the narrative, two figures, and the numbers behind them. Two slides are wanted — a **WHY** slide and a **WHAT + HOW** slide — qualitative, one graph each, no equations. Section 2 and section 3 are those two slides. The figures are vector, so they can be lifted at any size. **Do not invent numbers:** everything quotable is in section 5, and section 6 lists what must not be claimed.

1. The one thing this work fixes

The model produced an optimum. The people who fund these projects do not read optima — they read cash flows: money out, money in, year by year, cumulative, and the date it turns positive. Capex, opex, subsidies and CO_2 were all inside the model, but entangled in its equations, and nothing drew them against time.

That half step now exists. **The discounted cash flow equals the model's own NPV to the cent** — the same object, re-presented, checked on every build. Nothing was recalibrated to make it agree.

2. Slide one — WHY

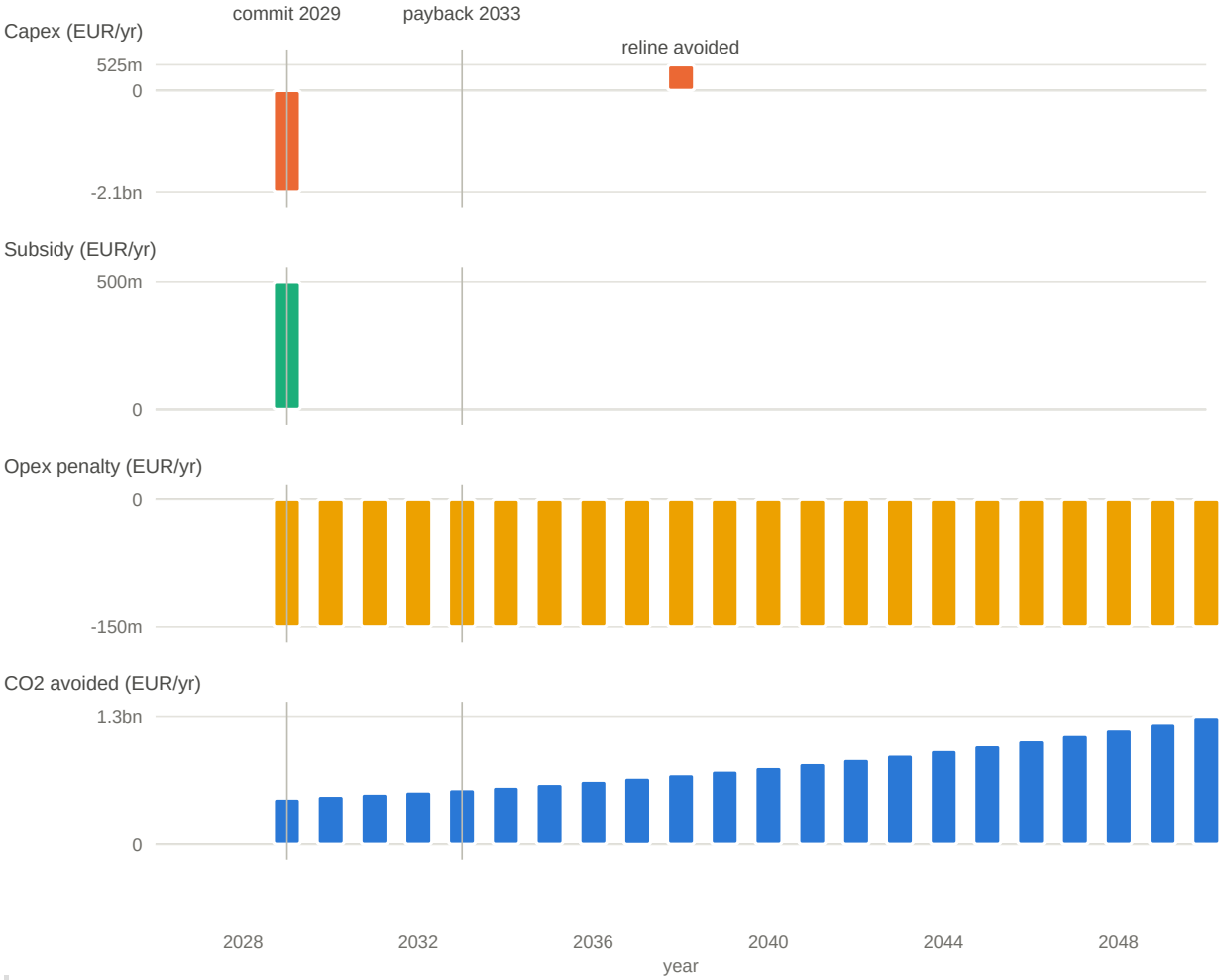
A multi-variable decision with no stable answer

Four inputs move independently, on different clocks, and the firm controls none of them. **Capex** falls with learning, but switching early forfeits undepreciated book value. **Opex** is dominated by the hydrogen price, the single largest swing factor. **Subsidies** are a window, not a pot — full weight to 2031, tapering to nothing by 2035. **CO_2** is a price that rises while free allocation phases out, and the two compound.

The honest framing: **there is no single answer, and there should not be.** Every plant has a different asset age, balance sheet, cost of capital and tolerance for non-productive time. What this produces is a map and a method.

The four inputs, as they actually fall over time

Capex, subsidy, opex and CO2 for a 2.5 Mt/yr plant committing in 2029. Bars throughout, because these are annual flows over discrete years. Each panel has its own scale - a EUR 2.1bn spike and a EUR 150m/yr charge do not share an axis usefully - and all four share the year axis.



> CAPEX is a spike, not a profile: the model commits in one year, so there is no construction schedule. SUBSIDY is paid against capex in that same year and is worth less the later it comes - full weight to 2031, then a taper to nothing by 2035. CO2 steepens because the carbon price rises AND free allocation phases out. OPEX IS FLAT, and that is the model's limitation rather than a finding: it carries one constant, where a real plant's operating cost rises with age, jumps at the switch while the technology is learned, then settles. Hole H10, comment 5.

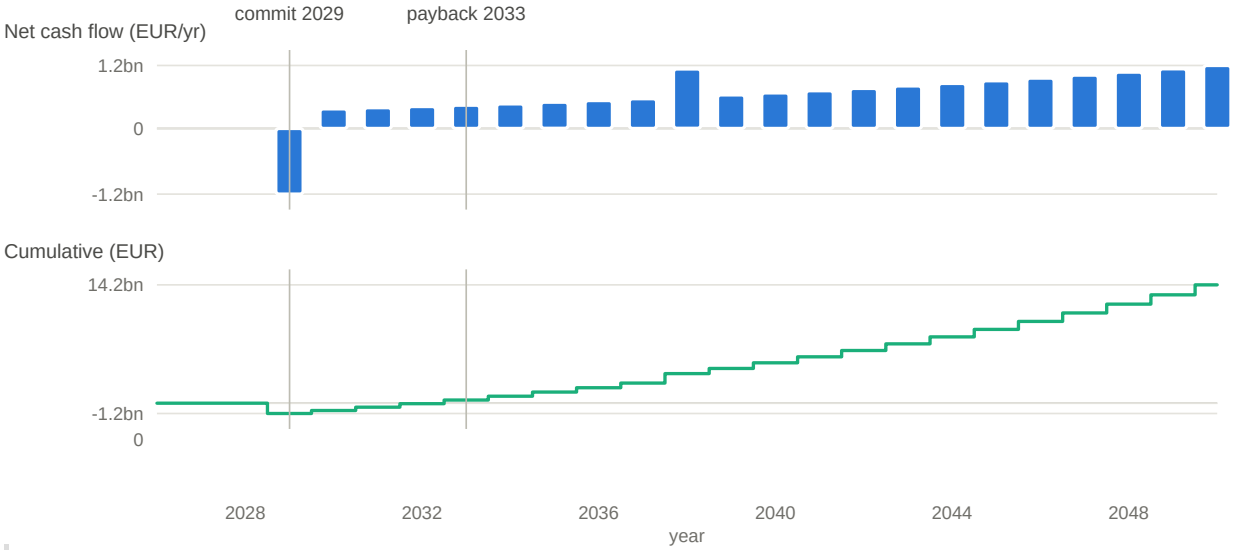
> Suggested for slide one. Four panels, shared year axis, each scaled to its own data — a €2.1bn spike and a €150m/yr line do not share an axis usefully. It makes the multi-variable point without a word of theory.

3. Slide two — WHAT and HOW

Translated into the language of a cash flow

The same decision as a cash flow

Money out and money in, year by year, then accumulated. This is the form debt is sized against - not an optimum, but a trough and a date.



> Peak exposure is -EUR 1.2bn in 2029: that is the financing requirement, and it is deeper than the capex alone because the cumulative keeps falling while the annual net is still negative. Cumulative turns positive in 2033 - 4 years after committing. Discounting this same series at 8% gives EUR 4.0bn, which is the model's own NPV(T*) of EUR 4.0bn - the two agree by construction, and analysis/verify.py checks it. Undiscounted and nominal-free: every figure is real EUR.

Commitment year (NPV-optimal)	**2029**
Peak exposure — the financing requirement	**-EUR 1.2bn in 2029**
Payback	**2033 — 4 years after committing**
NPV at that decision, published inputs	**EUR 4.0bn**
Carbon price that justifies committing	**€110/tCO2**

Two points that make this research rather than a result. **The answer is a range that crosses zero** — which end you believe is the whole argument, and that is uncertainty about *inputs*, separate from the price uncertainty the option machinery handles. **Timing is far more robust than value**: the corrections move the level of the answer much more than they move the year. The model is better at *when* than at *how much*.

4. The single strongest point, if only one lands

Realised volatility from the EU ETS auction record is **σ = 0.31** against the **0.20** the model runs on. Sigma is the only parameter that moves the *threshold* without moving the break-even, so it is the only live observation that changes *when*. At the specification's own ‘crisis’ rung (0.30) the trigger is €140 instead of €110, and the price path crosses it in **2035** rather than 2030 — 5 more years of waiting.

> The model still runs on 0.20. One year of auctions is not a price path to 2050, and nothing observed has been fed back into the parameters.

5. The numbers

5a. The cash flow, year by year — for a 2.5 Mt/yr plant

> Real euros, undiscounted. Money out is negative. Highlighted rows are the commitment year and the payback year. Discounting this series at 8% returns EUR 4.0bn, the model's NPV.

YEAR	CAPEX	SUBSIDY	OPEX	CO2 AVOIDED	NET	CUMULATIVE
2028	EUR 0	EUR 0	EUR 0	EUR 0	EUR 0	EUR 0
2029	**-EUR 2.1bn**	**EUR 500m**	**-EUR 150m**	**EUR 488m**	**-EUR 1.2bn**	**-EUR 1.2bn**
2030	EUR 0	EUR 0	-EUR 150m	EUR 518m	EUR 368m	-EUR 869m
2031	EUR 0	EUR 0	-EUR 150m	EUR 541m	EUR 391m	-EUR 479m
2032	EUR 0	EUR 0	-EUR 150m	EUR 564m	EUR 414m	-EUR 65m
2033	**EUR 0**	**EUR 0**	**-EUR 150m**	**EUR 588m**	**EUR 438m**	**EUR 373m**
2034	EUR 0	EUR 0	-EUR 150m	EUR 613m	EUR 463m	EUR 836m
2035	EUR 0	EUR 0	-EUR 150m	EUR 643m	EUR 493m	EUR 1.3bn
2036	EUR 0	EUR 0	-EUR 150m	EUR 675m	EUR 525m	EUR 1.9bn
2037	EUR 0	EUR 0	-EUR 150m	EUR 709m	EUR 559m	EUR 2.4bn
2038	EUR 525m	EUR 0	-EUR 150m	EUR 745m	EUR 1.1bn	EUR 3.5bn
2039	EUR 0	EUR 0	-EUR 150m	EUR 782m	EUR 632m	EUR 4.2bn
2040	EUR 0	EUR 0	-EUR 150m	EUR 821m	EUR 671m	EUR 4.8bn
2041	EUR 0	EUR 0	-EUR 150m	EUR 862m	EUR 712m	EUR 5.5bn
2042	EUR 0	EUR 0	-EUR 150m	EUR 905m	EUR 755m	EUR 6.3bn
2043	EUR 0	EUR 0	-EUR 150m	EUR 950m	EUR 800m	EUR 7.1bn
2044	EUR 0	EUR 0	-EUR 150m	EUR 998m	EUR 848m	EUR 8.0bn
2045	EUR 0	EUR 0	-EUR 150m	EUR 1.0bn	EUR 898m	EUR 8.8bn
2046	EUR 0	EUR 0	-EUR 150m	EUR 1.1bn	EUR 950m	EUR 9.8bn
2047	EUR 0	EUR 0	-EUR 150m	EUR 1.2bn	EUR 1.0bn	EUR 10.8bn
2048	EUR 0	EUR 0	-EUR 150m	EUR 1.2bn	EUR 1.1bn	EUR 11.9bn
2049	EUR 0	EUR 0	-EUR 150m	EUR 1.3bn	EUR 1.1bn	EUR 13.0bn
2050	EUR 0	EUR 0	-EUR 150m	EUR 1.3bn	EUR 1.2bn	EUR 14.2bn

5b. The answer as a range — four input sets

CASE	CAPEX I0	HYDROGEN	NPV	NPV, 2.5 MT/YR	TRIGGER	T*
base	**900**	**2.60**	**1,606**	**EUR 4.0bn**	**109.5**	**2029**
low	1,200	3.69	991	EUR 2.5bn	172.4	2031
central	**1,420**	**5.00**	**537**	**EUR 1.3bn**	**239.4**	**2038**
high	2,600	8.65	-124	-EUR 310m	460.4	none

> From the specification's published values down to ones taken from primary documents. The range crosses zero, so the sign of the answer is set by which inputs you accept, not by the model.

5c. The model against the live market

QUANTITY	ASSUMED	OBSERVED	RATIO	UNIT	FOR 2.5 MT/YR	AS OF
EUA price, 2026	76.00	84.02	1.11x	EUR/tCO2	EUR 38m	2026-08-24..2026-09-22
volatility sigma	**0.20**	**0.31**	**1.57x**	**1/yr^0.5**	**-**	**2026-01-08..2026-09-22**
Innovation Fund, YTD	EUR 500m	EUR 2.0bn	4.01x	EUR	EUR 500m	2026-01-07..2026-09-22
discount rate r	8.00	1.35	0.17x	% / yr real	-	2026-09-21
power vs gas, DE	0.04	0.14	3.86x	EUR/kWh	-	2025-S2

> Source: EEX EU ETS primary-market auction report, read at build time. Observed 2026-08-24..2026-09-22. Nothing here feeds the model.

5d. Context figures

Capex assumption I0	**€900/t capacity = EUR 2.2bn**
— vs every project actually under construction	**€1,300+/t**
Support s_max	**€200/t capacity = EUR 500m**
EU ETS auction revenue, 2026 year to date	**EUR 27.5bn**
— of which the Innovation Fund	**EUR 2.0bn (25% would go to one plant)**
Discount rate	**8% real, to 2026**
Horizon	**2050**

6. What must NOT be claimed

State these before anyone asks. They are the credibility of the work.

- **Opex has no time shape.** The model carries one constant driven only by the hydrogen price. A real plant's operating cost rises with age, jumps at the switch while an unfamiliar technology is learned, then settles lower as contracts improve. None of that is modelled — the line is drawn flat and labelled flat rather than given a plausible curve. Open hole H10.
- **No tax layer.** Electricity is 2–4× gas per kWh in every member state, and those prices exclude taxes, levies and network charges — exactly what “fiscal harmonisation” would change.
- **No MATLAB reference.** The 50-case diff harness is built and its negative control works; there is no reference file to run it against.
- **Ladder rows 4–7 do not reproduce** the published table. Documented case by case in `CALIBRATION.md`.
- **One unverified input** — the €8.29/kg Platts hydrogen assessment, licensed data taken from the specification.
- **Subsidy money is not earmarked.** The ETS Directive's spending clause is a non-binding target on national budgets, not an entitlement any project can draw on.
- **This is an increment, not a P&L.** The cash flow is the difference between switching and not switching. Steel revenue, raw materials, debt service, tax and working capital net out by construction. A real sizing exercise starts here and adds the rest.

7. What is checked

- **Validation gate.** Nine published quantities reproduce to 1e-3 relative before anything downstream prints.
- **Every table cell re-derived** from the raw CSV or cached statistic — not from the function that prints it — and compared at the precision shown. **145/145 agree.**
- **The specification, quantity by quantity.** 54 of 62 published figures reproduce; all eight differences are NPV *levels*, high by 1.1–8.0%. No year, trigger or crossing year differs.
- 368 tests and 42 standard-library self-checks on every build; zero third-party dependencies.
